STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-PRO-ISOMORPHIC

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6451-6459 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 6461-6504, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_sdga.json (pb=1330, stmt 6451-6459, proof 6461-6504).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0G2W: sources/stacks-sdga/pinned-original-source.tex, source lines 6504–6504. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above, suppose that $\mathcal{A} = (A_n)$ and $\mathcal{B} = (B_n)$ are inverse systems of differential graded R-algebras. If $\varphi : (A_n) \to (B_n)$ is an isomorphism of pro-objects, then the functor $QC(\mathcal{A}) \to QC(\mathcal{B})$ constructed above is an equivalence.*

Proof. Let $\psi : (B_n) \to (A_n)$ be a morphism of pro-objects which is inverse to φ. According to the discussion in Categories, Example 0G2W we may assume that φ is given by a system of maps as above and ψ is given $n(1) < n(2) < \ldots$ and a commutative diagram

$$\begin{array}{ccccccc} \ldots & \longrightarrow & B_{n(3)} & \longrightarrow & B_{n(2)} & \longrightarrow & B_{n(1)} \\ & & \downarrow{\scriptstyle \psi_3} & & \downarrow{\scriptstyle \psi_2} & & \downarrow{\scriptstyle \psi_1} \\ \ldots & \longrightarrow & A_3 & \longrightarrow & A_2 & \longrightarrow & A_1 \end{array}$$

of differential graded R-algebras. Since $\varphi \circ \psi = \mathrm{id}$ we may, after possibly increasing the values of the functions $n(\cdot)$ and $m(\cdot)$ assume that $B_{n(m(i))} \to A_{m(i)} \to B_i$ is the transition map. It follows that the composition of the functors

$$QC(\mathcal{B}) \to QC(\mathcal{A}) \to QC(\mathcal{B})$$

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

sends a good sheaf of differential graded $\mathcal{B}$-modules $\mathcal{N} = (N_n)$ to the inverse system $\mathcal{N}' = (N'_i)$ with values

$$N'_i = N_{n(m(i))} \otimes_{B_{n(m(i))}} B_i$$

which is canonically quasi-isomorphic to $\mathcal{N}$ exactly because $\mathcal{N}$ is an object of $QC(\mathcal{B})$ and because N_j is a K-flat differential graded module for all j. Since the same is true for the composition the other way around we conclude. $\square$